

BENCHMARK: NON-DETECTION

Results

|                                                   |                                   |
|---------------------------------------------------|-----------------------------------|
| Mid-Transit Time (Tmid)                           | 2460828.93413 +/- 0.00092 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.18 +/- 0.053                    |
| Transit depth (Rp/Rs)^2                           | 3.23 +/- 1.91 [%]                 |
| Orbital Inclination (inc)                         | 89.87 +/- 1.89                    |
| Airmass coefficient 1 (a1)                        | 4.16 +/- 2.1                      |
| Airmass coefficient 2 (a2)                        | -1.48 +/- 0.44                    |
| Scatter in the residuals of the lightcurve fit is | 48.23 %                           |
| Best Comparison Star                              | None                              |
| Optimal Aperture                                  | 4.18                              |
| Optimal Annulus                                   | 5.21                              |
| Transit Duration (day)                            | 0.094 +/- 0.019                   |

Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                          |
|---------------------------|------------------------------------------|
| Rp/R■ fit vs published    | 0.18 ± 0.053 vs 0.1356 (deviation 0.55σ) |
| Duration fit vs published | 0.094 d vs 0.1129 d (deviation 0.65σ)    |
| Mid-time O-C              | -1.8 min                                 |
| Depth SNR                 | 2.2                                      |
| Residual scatter          | 48.23 %                                  |

Light curve

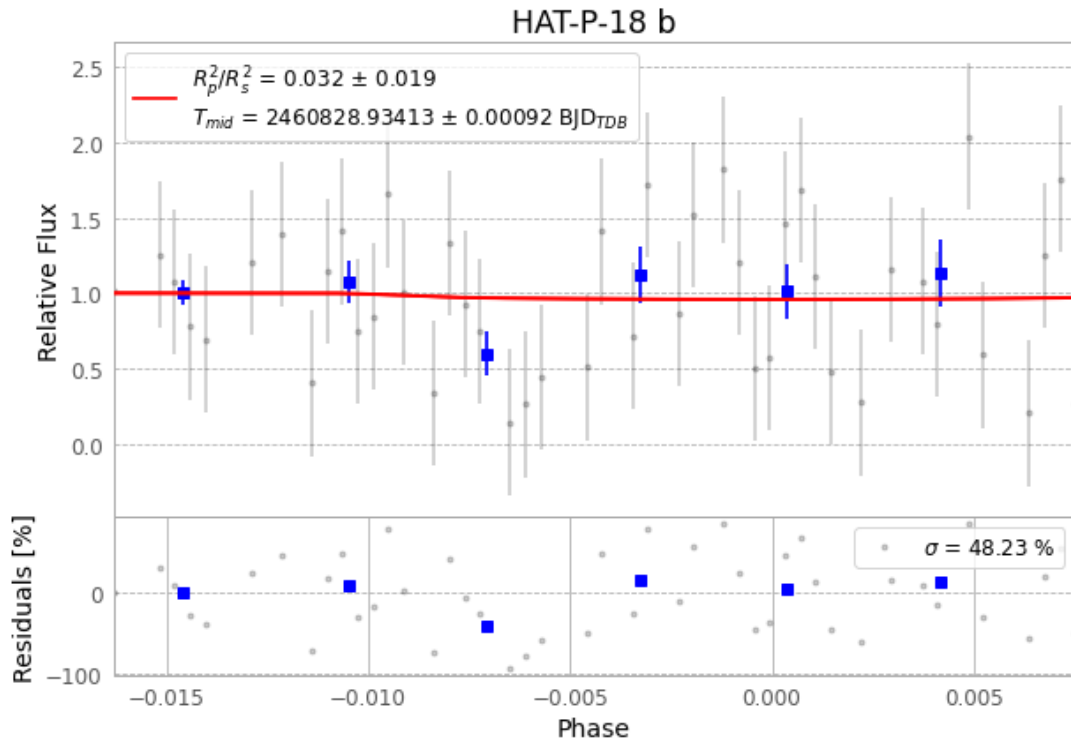


Figure 1 — FinalLightCurve\_HAT-P-18 b\_2025-06-02.png (SHA-256 0e879b174be002b1...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [475, 156], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 983c2ac0b1d543dc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 1dc3d22

## Data provenance

64 FITS frames (42 MB), HATP-18250602080915.FITS → HATP-18250602111815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-18-250602.log (95c1c9ccc35a...), FinalLightCurve\_HAT-P-18 b\_2025-06-02.png (0e879b174be0...), FinalLightCurve\_HAT-P-18 b\_2025-06-02.csv (417a2b46970e...)

## Compute resources

Wall time 195.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 00:31Z.